toys10_04-beat-runner_ORDERING

04 Beat Runner: ordering guide (what to buy, where, in what order)

> Companion to `MANUFACTURING_MANUAL.md`. Prices and stock are from LCSC on the run date; vendors, shipping and customs rules change: verify before paying. General rules and vendor list: `manuals/ORDERING_GUIDE.md`.

1. Order of work

1. Read the manual and the caveat above.
2. Order the circuit boards and SMD assembly (section 2).
3. While the boards travel, order the other items (sections 3 and 5) and make the printed or laser-cut parts.
4. Assemble, program and test only after everything has arrived; follow the manual.

2. Order the circuit boards (JLCPCB, PCB + assembly)

Files (all in this repository):

- Gerbers: `utility10/pcb/release/beatrunner/1_fabrication/beat_runner_gerbers.zip`
- Assembly BOM: `utility10/pcb/release/beatrunner/2_assembly/beat_runner_JLC_BOM.csv`
- Pick-and-place (CPL): `utility10/pcb/release/beatrunner/2_assembly/beat_runner_JLC_CPL.csv`
- Full BOM with LCSC numbers and prices: `costing/beatrunner_priced_BOM.csv`

Steps:

1. Create an account at **jlcpcb.com** and click *Order now*. Upload the Gerber ZIP. The viewer must show the board outline, both copper layers, silkscreen and drill holes.
2. Settings: **2 layers, 1.6 mm, FR-4**, lead-free HASL (use **ENIG** if the board has an LGA/QFN fine-pitch part, e.g. ADXL345, ESP32 module pad, BQ25570), green mask, white silk, quantity 5 (the minimum), *Remove order number: specify a location* if you want to avoid the order-number mark. Heavier copper (2 oz) only where the board notes say so.
3. Switch on **PCB Assembly**: *Economic*, assembly side *Top*, and set tooling holes *Added by JLCPCB* unless the board already has them. Choose the number of boards to assemble (2 or 5).
4. Upload the BOM and CPL. In the parts step, every line must match (LCSC numbers are in the BOM). Where a part is shown as *out of stock* or *no match*, use the alternative suggested by JLCPCB only after checking footprint and value against the list below; do not accept a different footprint.
5. In the **component placement preview**, check pin 1 / polarity of every IC, diode, LED, electrolytic capacitor, USB-C and connector against `*_assembly_top.pdf` in the same folder. Wrong rotation is the most common assembly error: ask JLCPCB support to rotate before approving.
6. Pay, then approve the assembly engineering review messages (they usually ask about a part or rotation). Delivery by DHL/FedEx/UPS takes about 1 to 2 weeks to India.

Parts **not** assembled by JLCPCB (you buy and solder them yourself):

Refs	Value	Part	LCSC
J2	ISP	HCTL PZ254-2-03-Z-8.5 (ZHOURI C5116479 had only 340 in stock)	C2894964
J1	BATT 3xAAA	JST B2B-PH-K-S(LF)(SN)	C131337
BZ1	Buzzer	YUEXIN KC1206 16 ohm passive magnetic 12x9.5 mm (2048 Hz)	C781869
J3	Arcade button	JST B2B-PH-K-S(LF)(SN)	C131337
TP1, TP2	TP		

Order these with the same LCSC order (add them to the cart so one shipment arrives) or buy locally in India (section 5).

Parts cost and stock (live LCSC prices on the run date)

Total parts per board: **US4.48*for1board* * *, * * *US*3.96 each for 10 boards**, **US$3.32 each for 100 boards** (about INR 394 / 348 / 292 at 88 INR/USD). The per-line break-up is in `costing/beatrunner_priced_BOM.csv`. This is parts only: PCB fabrication, assembly fees, shipping, customs and GST come on top (see `manuals/ORDERING_GUIDE.md`, section 6).

Stock warnings on the run date:

- TP (no lcsc price (hand-fit or not priced), stock)

Refs	Qty	Value	Part	LCSC	Unit USD (1 board)
U1	1	ATmega328P-AU	ATMEGA328P-AU	C14877	2.3308
C1,C2,C3,C5,C6	5	100n	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	0.0075
R7	1	10k	0603 10k 1%	C25804	0.0024
J2	1	ISP	HCTL PZ254-2-03-Z-8.5 (ZHOURI C5116479 had only 340 in stock)	C2894964	0.0766
J1	1	BATT 3xAAA	JST B2B-PH-K-S(LF)(SN)	C131337	0.0359
SW1	1	Power	C&K PCM12SMTR	C221841	1.2754
Q2	1	AO3401A	AO3401A	C15127	0.0969
C4	1	10u	0805 10uF X5R	C15850	0.0441
BZ1	1	Buzzer	YUEXIN KC1206 16 ohm passive magnetic 12x9.5 mm (2048 Hz)	C781869	0.1513
Q1	1	MMBT3904	MMBT3904	C20526	0.0118
D2	1	1N4148WS	1N4148WS	C2128	0.0120
R6	1	1k	0603 1k 1%	C21190	0.0017
U2,U3	2	74HC595	74HC595D	C5947	0.1079
J3	1	Arcade button	JST B2B-PH-K-S(LF)(SN)	C131337	0.0359
TP1,TP2	2	TP			
D10,D11,D12,D13,D14,D15,D16,D17,D18,D19,D20,D21,D22,D23,D24,D25	16	Red	Hubei KENTO KT-0603R red (Everlight 19-217 was out of stock)	C2286	0.0077
R20,R21,R22,R23,R24,R25,R26,R27,R28,R29,R30,R31,R32,R33,R34,R35	16	330	UNI-ROYAL 0603WAF3300T5E	C23138	0.0021

2b. Enclosure: printed parts and fasteners (generated from the board)

- Print `4_enclosure/` base and lid (outer about 165 x 108 x 60 mm) in PETG or ABS, 3 perimeters, 25 % infill; the lid closes with **4 x M2.5 x 10 mm self-tapping screws** into the corner columns; the board sits on its standoffs with 4 x M2.5 x 6 mm self-tapping screws.
- Battery: **Keystone 2480 (3 x AAA, wire leads)** in the floor bay (54.7 x 39.5 x 13.8 mm); print `*_enclosure_battery_door.stl`; the door closes with 2 x M2.5 x 6 mm self-tapping (tool needed to open).

Route the holder lead through the hole in the bay ceiling to the battery connector.

- Print `*_enclosure_caps.stl` (stalk): stalk for SW1. Light pipes in clear PETG or resin; fit plungers and pipes from inside before closing the lid.
- power switch: printed stalk through a lid slot; one 30 mm arcade button and a window over the 16 LEDs (glue a 1 mm translucent acrylic strip inside); 3 x AAA bay with a screwed door
- Print one set first and test-fit the board, the holder and every cap; the dimensions are first-pass.

3. Bought-in and mechanical items

3b. Bought-in items

Item	Rs
MCU + crystal	55
PCB	35
2 × 74HC595	14
16 LEDs + resistor arrays	30
Arcade button	15
Buzzer	12
Battery/USB power	30
Enclosure	60
Packaging + manual	30
Assembly + test	45

- One 30 mm arcade button (J3), 3 x AAA holder + switch, enclosure, 16 x red 0603 LEDs are on the board.

4. Confidence (read before you order)

- **Order confidence: Medium-High**: limited stock on: BZ1 Buzzer (stock 140).
- **Works-as-intended confidence: Medium-High**: Rhythm game on its own board; logic host-tested; LEDs and buzzer circuits are standard.
- **First thing to do:** Check the LED brightness and button spacing.
- Rubric and the reasons for every project: `costing/CONFIDENCE.md`. These are judgements, not measurements: nothing here has been built.

5. Where to buy and whom to approach

Item type	India (fast, GST invoice)	Overseas (cheaper, 1 to 3 weeks, customs)
Printed circuit board and SMD assembly	JLCPCB is the practical choice for 5 to 100 boards; local PCB shops quote per project (ask for the Gerber ZIP price)	**JLCPCB** (jlcpcb.com), PCBWay (pcbway.com)
Connectors, headers, passives, ICs by LCSC number	element14 India, Mouser India, DigiKey India (genuine, GST invoice, higher price); local electronics markets for common parts	**LCSC** (lcsc.com) with the same cart as JLCPCB
Modules: ESP32 dev boards, SIM7080G, LoRa (RA-08 / RAK3172), pH / EC / DO modules, relay modules, load cells, ultrasonic sensors	Robu.in, Robocraze, ThinkRobotics, Sunrom Electronics, Amazon.in	AliExpress, Seeed Studio, Waveshare
Motors, stepper drivers, wheels	Robu.in, Robocraze, local robotics shops	StepperOnline (omc-stepperonline.com)
Li-ion / LiFePO4 cells and packs, solar panels, MPPT controllers	Authorised battery and solar dealers, Amazon.in (buy protected cells with an invoice)	not recommended by post (shipping rules for batteries)
Laser-cut and 3D-printed parts	Local laser-cutting and 3D-printing shops (search IndiaMART, Justdial or maker spaces): send the SVG/DXF/STL from the repository and ask for a quote per piece	JLC3DP (jlc3dp.com), PCBWay 3D printing

Item type	India (fast, GST invoice)	Overseas (cheaper, 1 to 3 weeks, customs)
Fasteners, tubing, hardware, enclosures	Local hardware, Robu.in, Amazon.in	n/a

How to approach a vendor: send the project name, the exact file (BOM, CPL, Gerber ZIP, STL or SVG), the quantity, and ask for the price, the lead time, the GST invoice and the warranty or return terms. Keep the reply for your records.